Face Recognition Attendance System – Technical Roadmap & Server Build

Overview

This roadmap outlines the implementation plan for the Face Recognition Attendance System, detailing weekly tasks, milestones, and resources required for the project. It also includes a server build guide and workflow explanation.

Week 1: System Design & Setup

Objectives: - Finalize architecture and components - Set up development environment

Tasks: - Design final architecture (tablet → server → dashboard) - Set up GitHub repo and branch structure for collaboration - Configure Python environment (FastAPI, OpenCV, face_recognition) - Install and configure PostgreSQL/SQLite - Select tablet/device for testing

Resources: - Python 3.11+, FastAPI documentation - OpenCV / face_recognition tutorials - PostgreSQL docs - GitHub Collaboration guide

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Week 2: Teacher Biometric Login Module

Objectives: - Implement teacher face recognition login on tablets

Tasks: - Capture teacher faces for DB enrollment - Implement liveness check (blink + head turn) - Connect tablet camera feed to server - Verify teacher embeddings against DB - Display session feedback on tablet

Milestones: - Teacher can login via face recognition - Tablet shows session active feedback

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Week 3: Student Attendance Module

Objectives: - Implement student check-in with liveness and duplicate detection

Tasks: - Capture student faces for DB enrollment - Implement student liveness verification - Server-side recognition and embedding match - Duplicate attendance check per class per day - Tablet feedback for success / already checked

Milestones: - Students can check-in reliably - Duplicate attempts rejected

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Week 4: Admin Dashboard & Reporting

Objectives: - Implement web dashboard for teachers and admins

Tasks: - Dashboard login & teacher management - Real-time attendance view per class - Audit log view (hash-chain verification) - Class-wise attendance summaries - Device status monitoring

Milestones: - Admin can view all attendance data - Reports are exportable (CSV/PDF)

Week 5: Server Integration & Testing

Objectives: - Connect all modules, perform end-to-end testing

Tasks: - Integrate tablet, server, and dashboard modules - Test teacher login, student check-ins, feedback - Validate liveness detection and duplicate prevention - Test multiple tablets simultaneously - Document system flow

Server Workflow: 1. Teacher Login → Face capture, liveness check, embedding verification, session activation 2. Student Check-In → Face capture, liveness check, embedding verification, duplicate check, hash-chain logging 3. Admin Dashboard → Fetch attendance, teacher sessions, and device status

Resources: - Python, FastAPI, async programming tutorials - OpenCV / face_recognition guides - PostgreSQL / SQLite documentation

Milestones: - Teacher login, student check-in, and admin dashboard connected - End-to-end functionality working

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Week 6: Optimization & Presentation Prep

Objectives: - Optimize system performance, finalize documentation and demo

Tasks: - Reduce recognition latency (<500ms per student) - Optimize liveness verification - Polish UI / tablet feedback - Prepare slides and demo scenario - Create full documentation including architecture diagram, workflow, and roadmap

Milestones: - Fully functional system, demo-ready - PDF documentation complete

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Collaboration Notes

- GitHub branches: teacher-login, student-checkin, admin-dashboard
- Assign tasks to team members weekly

- Daily commits and stand-ups recommended
- Maintain README and code comments

Optional Enhancements

- Two-factor teacher authentication (password + biometric)
- Tablet GPS/location verification
- Deepfake detection for improved liveness
- Automated attendance export (CSV/PDF/email)

Goal: By following this roadmap, the system will be secure, realistic, fully functional, and presentation-ready, with clear weekly tasks and server integration guidance.